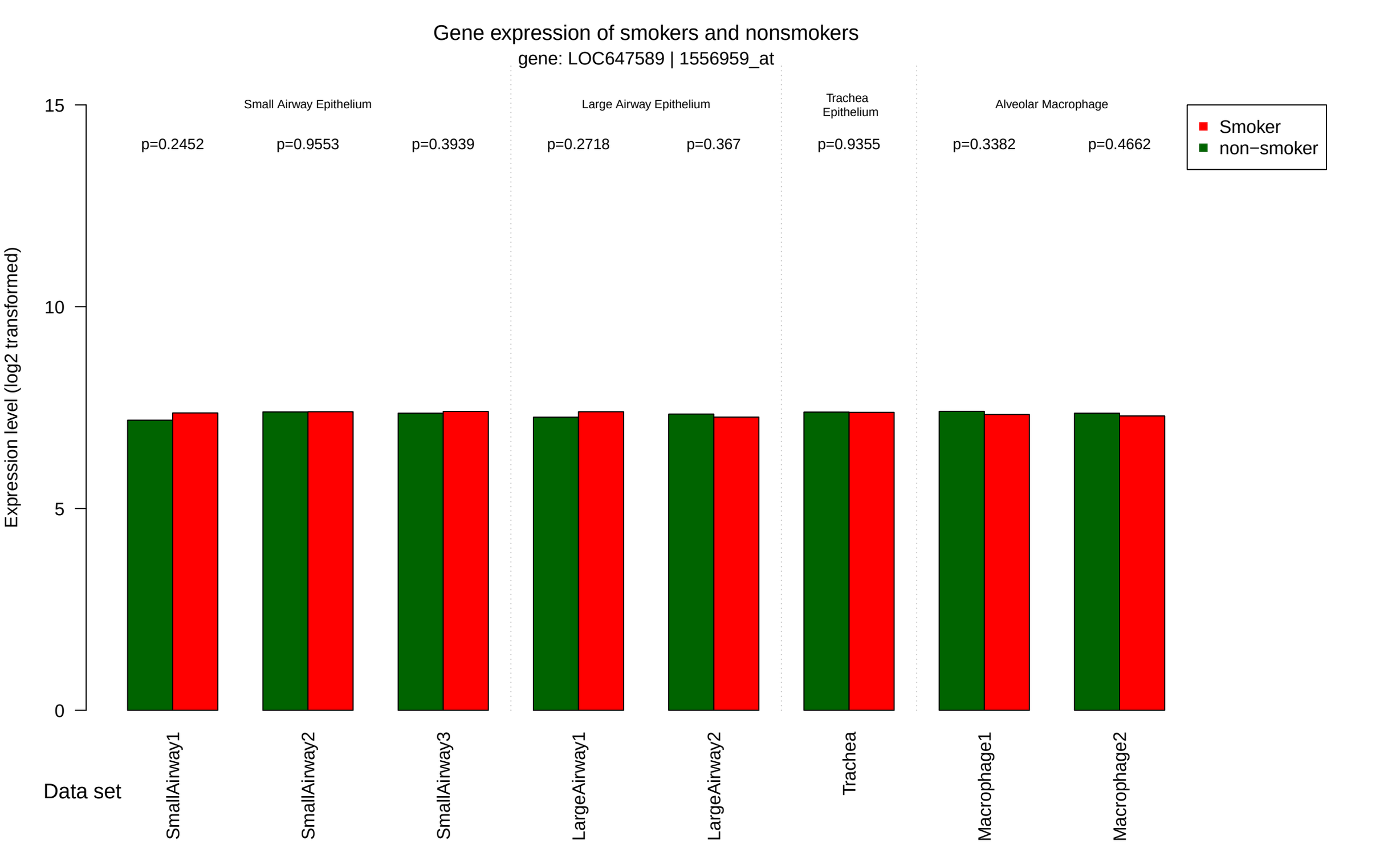
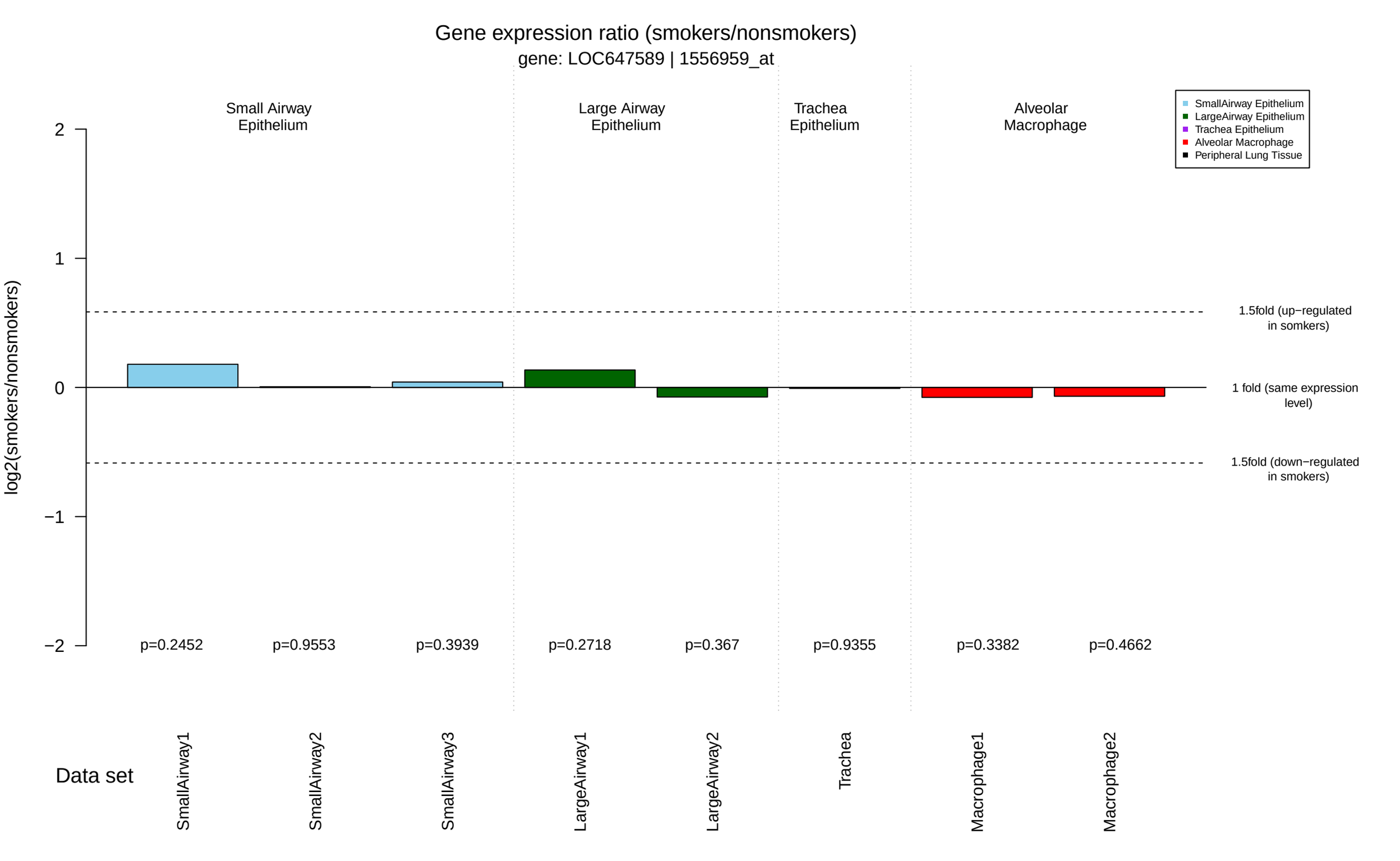
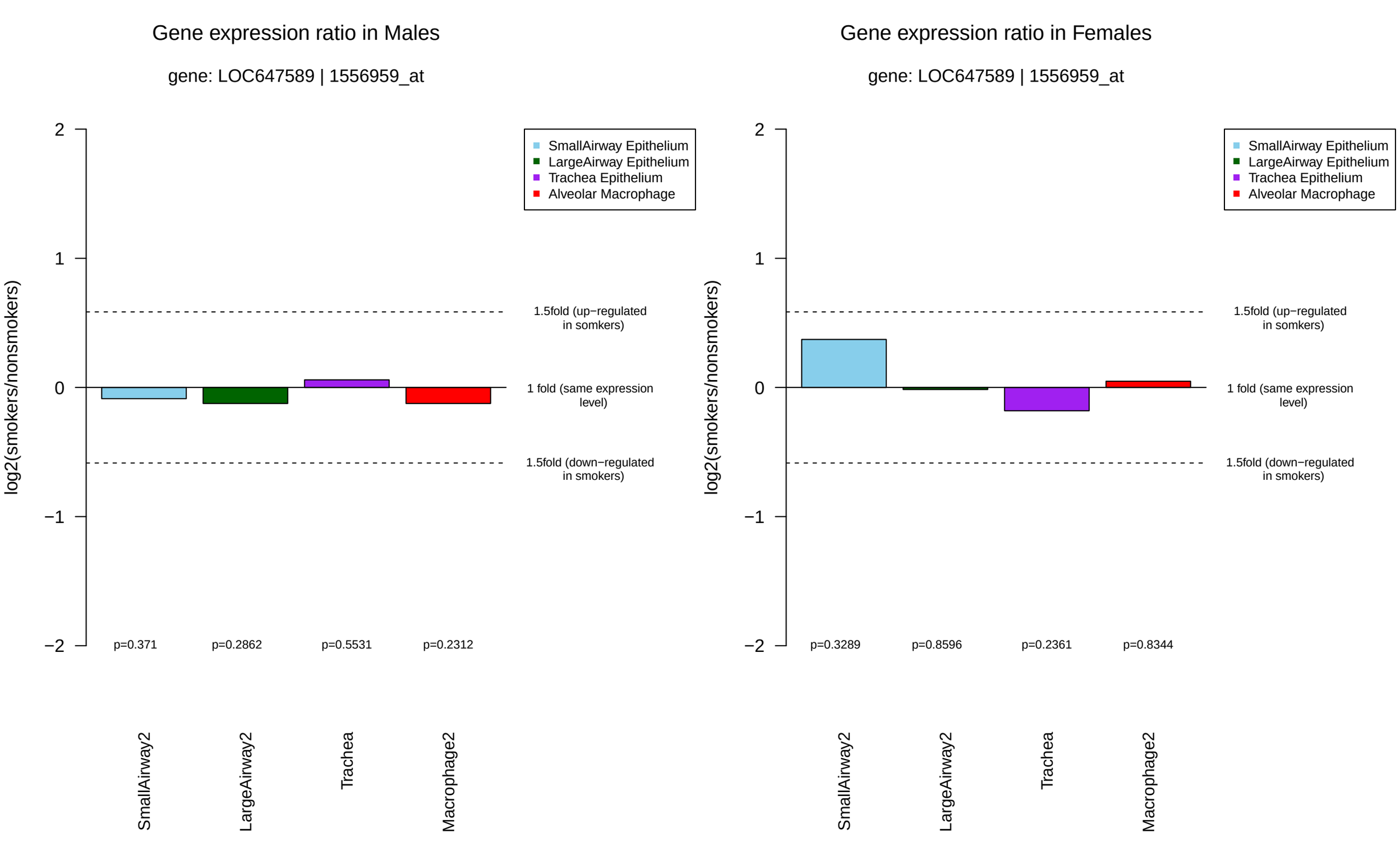
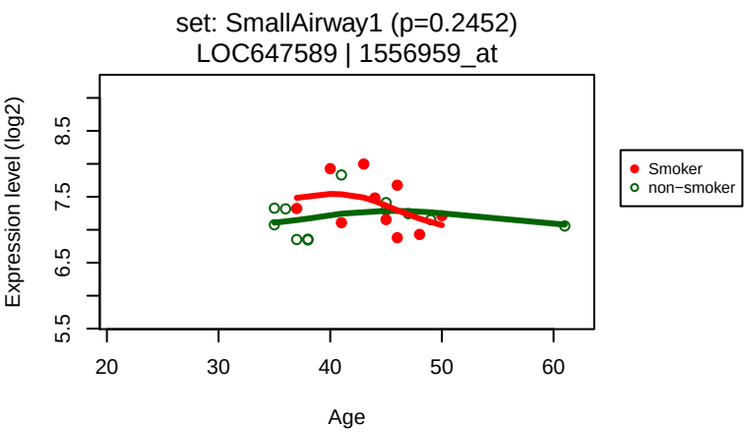
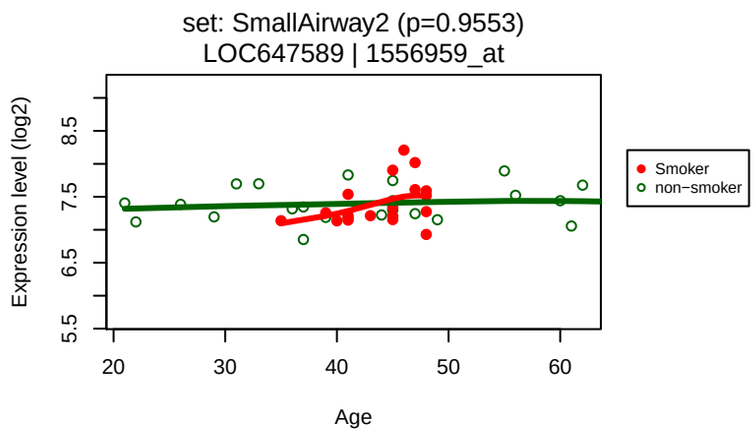
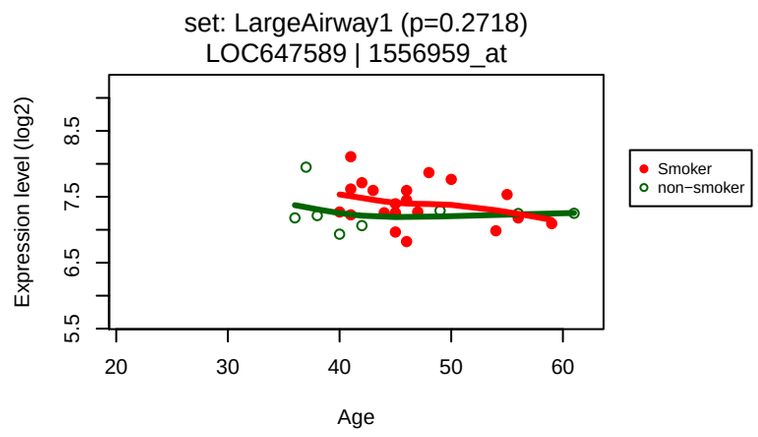

Gene expression of smokers and nonsmokers
gene: LOC647589 | 1556959_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2452
p=0.9553
p=0.3939
p=0.2718
p=0.367
p=0.9355
p=0.3382
p=0.4662
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

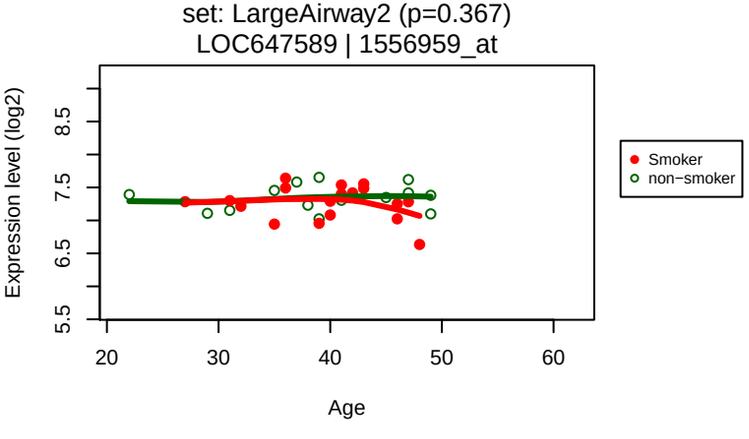

Gene expression ratio (smokers/nonsmokers)
gene: LOC647589 | 1556959_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2452
p=0.9553
p=0.3939
p=0.2718
p=0.367
p=0.9355
p=0.3382
p=0.4662
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

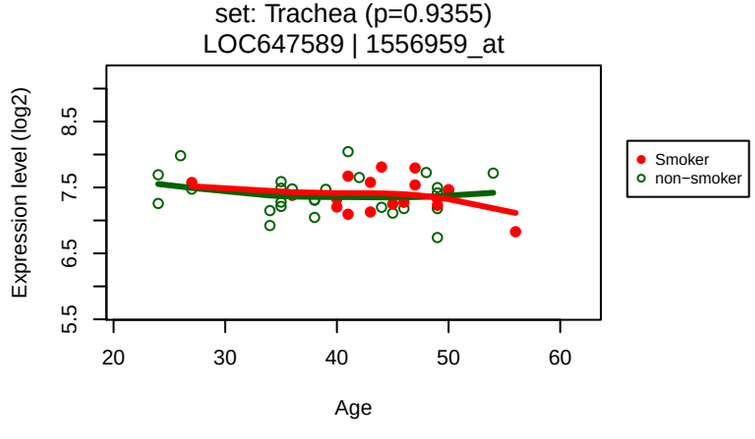

Gene expression ratio in Males
gene: LOC647589 | 1556959_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.371
p=0.2862
p=0.5531
p=0.2312
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC647589 | 1556959_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3289
p=0.8596
p=0.2361
p=0.8344
SmallAirway2
LargeAirway2
Trachea
Macrophage2

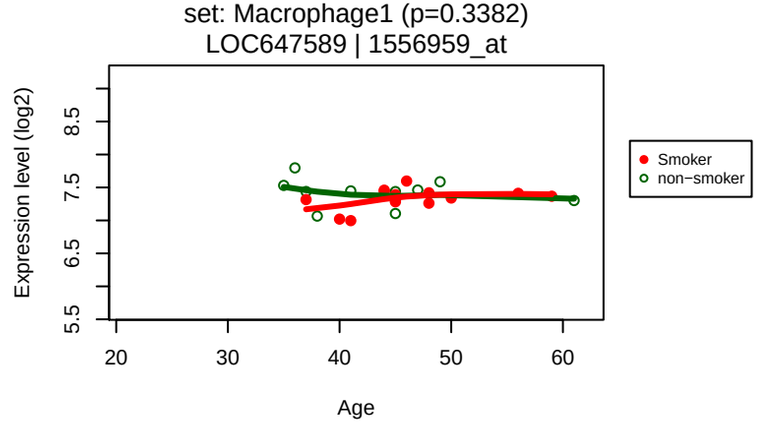

set: SmallAirway1 (p=0.2452)
LOC647589 | 1556959_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker

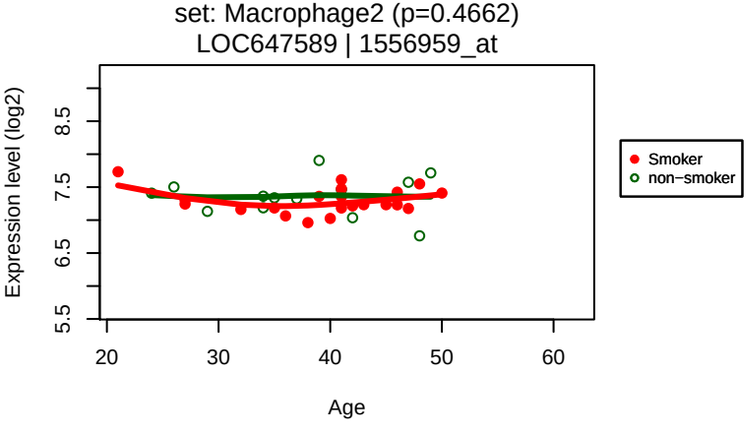

set: SmallAirway2 (p=0.9553)
LOC647589 | 1556959_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker

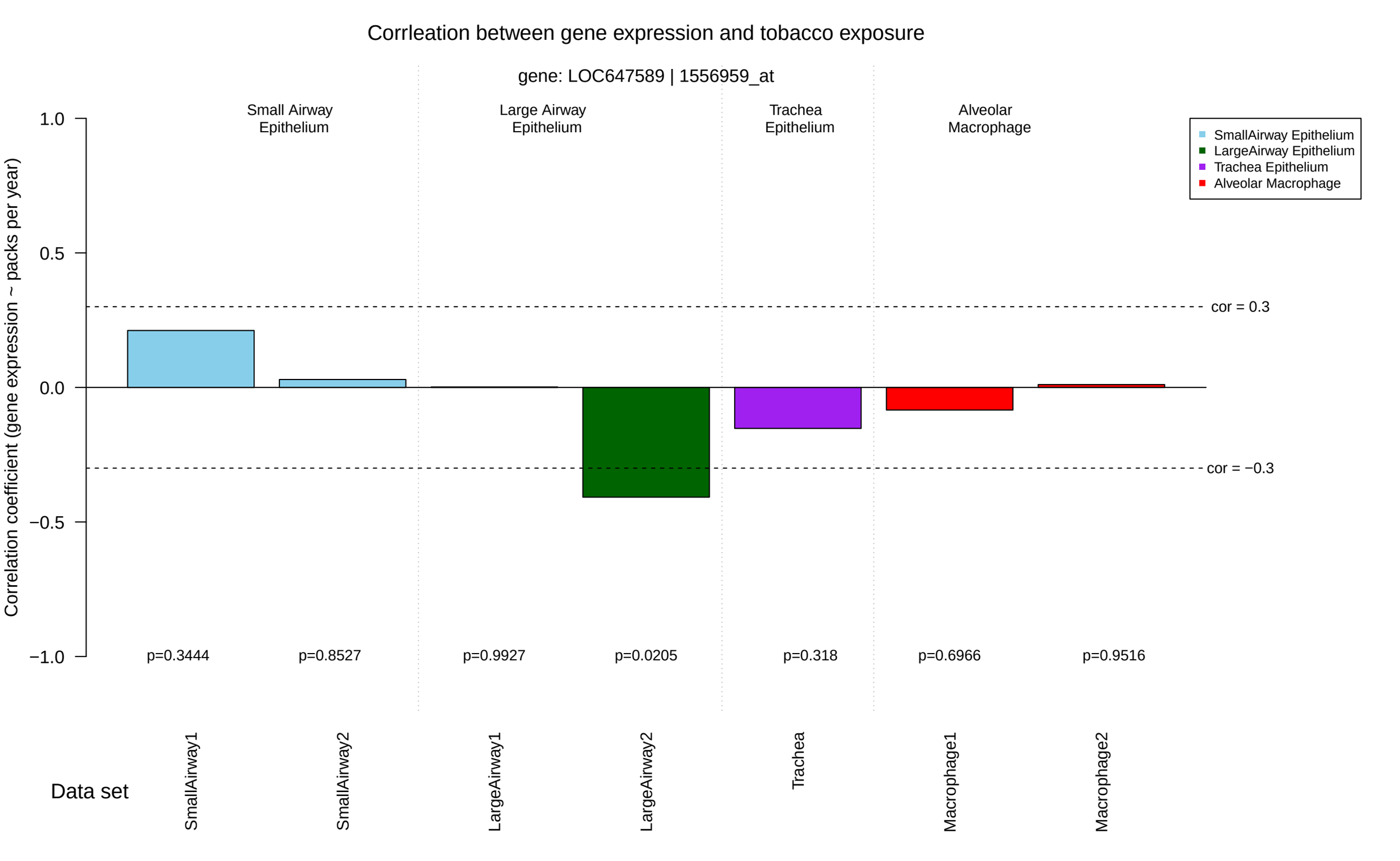

set: LargeAirway1 (p=0.2718)
LOC647589 | 1556959_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.367)
LOC647589 | 1556959_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.9355)
LOC647589 | 1556959_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3382)
LOC647589 | 1556959_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4662)
LOC647589 | 1556959_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC647589 | 1556959_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3444
p=0.8527
p=0.9927
p=0.0205
p=0.318
p=0.6966
p=0.9516
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2